

Polynomials degree stress prototype: Orientation map

Find degree after like terms cancel

Subject: Mathematics

Topic: Polynomials

Concept: Find degree after like terms cancel

Concept difficulty: Medium concept

Hard transition

Degree belongs to the simplified polynomial, so a written high-power term whose total coefficient is zero does not decide it.

power 0

power 1

power 2

power 3

Stage 1 of 3

Show powers 0, 1, 2 and 3 as coefficient slots before selecting the highest nonzero one.

Governing relation

$\text{degree}(p) = \text{largest } n \text{ with } a_n \text{ not equal to } 0 \text{ for nonzero } p(x) = \sum a_n x^n$

The degree is controlled by the highest power whose coefficient survives after simplification.

- Combine like powers first.
- At least one coefficient is nonzero; this prototype does not assign a degree to the zero polynomial.

Compact anchor

Degree of $3x^3 - 3x^3 + 2x + 5$?

The cubic terms cancel; $2x + 5$ has degree 1.

Before you go on

What is the degree of $4x^2 - 4x^2 + 7$? Explain the cancellation.